Hedging the Hype

An Advanced EDA of Big Tech Concentration and Macro Fragility

11-Chart Deep Dive | Magnificent 7 | Full Stack Analysis

Executive Abstract

This report extends the foundational EDA of the Magnificent 7 into a full quantitative risk framework. Where an exploratory analysis tracks price levels and broad correlations, this work examines WITHIN-basket stock dispersion, regime-conditional return distributions, rolling risk-adjusted quality (Sharpe decay), peak-to-trough drawdown profiles, individual stock rate sensitivity, and the evolution of intra-basket co-movement across four distinct market eras. Together, these 11 charts constitute a practitioner-grade stress test of the Mag 7 concentration thesis.

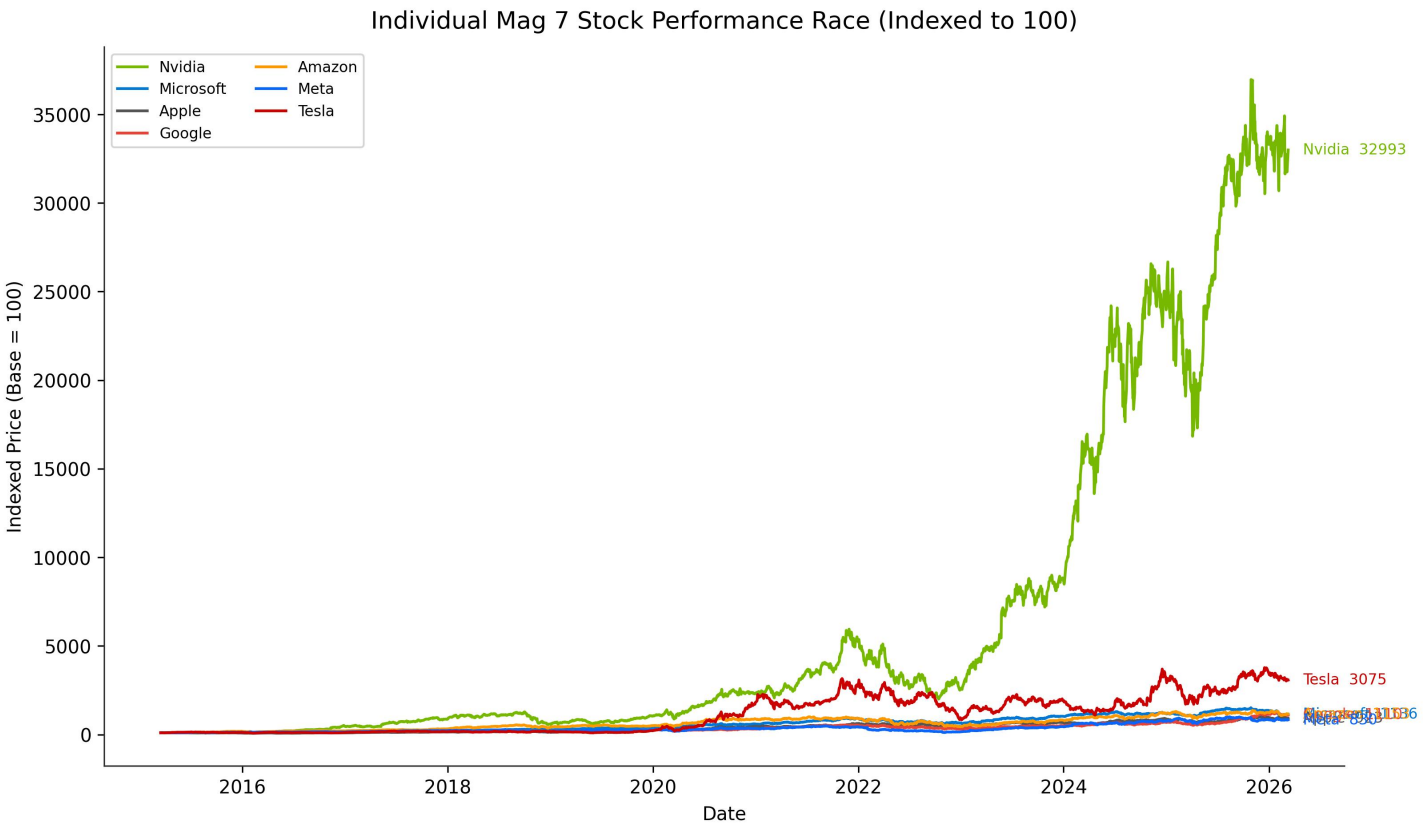
Key Quantitative Metrics

Mag 7 Rolling Sharpe (12M):	**1.23**
S&P 493 Rolling Sharpe (12M):	**0.26**
Total Mag 7 Return (Base=100):	**+6207.7%**
Macro Stress Days Identified:	**344.0**

Methodology & Feature Engineering

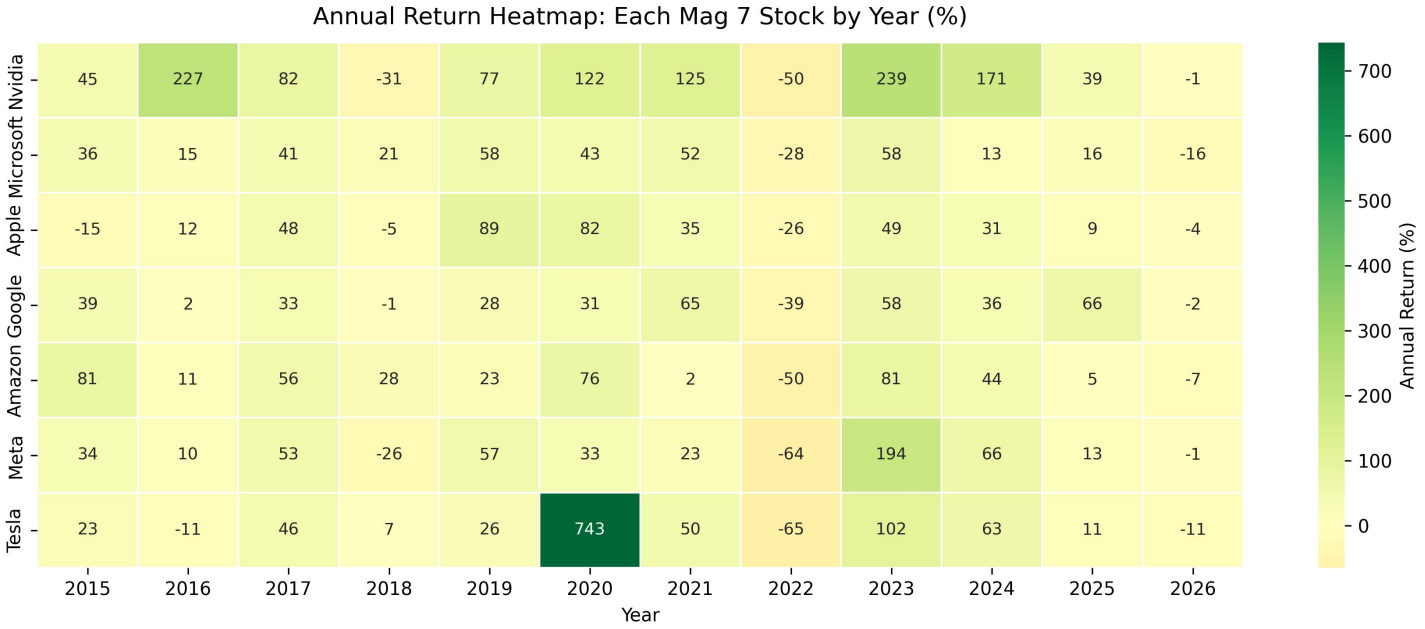
1. Data: magnificent_7_macro_nexus_supercharged.xlsx, parsed and forward-filled.
2. S&P 493 Proxy: Derived as SP493 = (SP500 - 0.25 x Mag7) / 0.75.
3. Normalisation: All stock prices indexed to 100 at series start.
4. Rolling Sharpe: 252-day window, excess returns over 13W T-Bill daily equivalent.
5. Drawdown: Peak-to-trough computed on cumulative return index.
6. Beta: Rolling 90-day OLS covariance-to-variance ratio vs S&P 500 returns.
7. Regimes: Yield curve bucketed into 4 bands; VIX bucketed into 4 fear levels.
8. Era Correlations: Returns vs macro pct-changes in 4 sub-periods.
9. Intra-basket Correlation: Mean upper-triangle of 7x7 rolling pairwise corr matrix.
10. Yield Sensitivity: OLS regression of each stock daily return vs 10Y yield daily change.

Chart A: Individual Stock Performance Race (Indexed to 100)



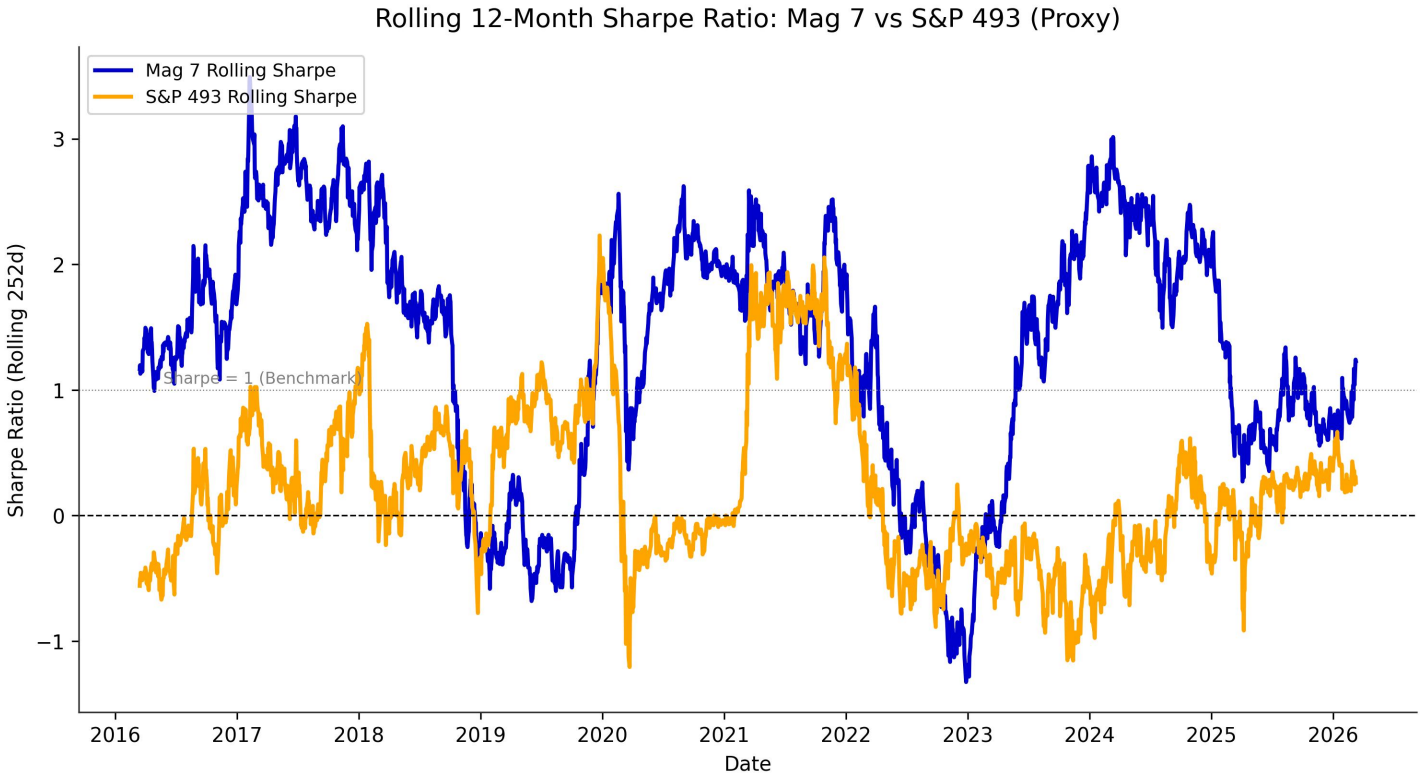
The Mag 7 is not a monolith. While the composite index grew dramatically, the within-basket dispersion is extreme. Nvidia outperformed every other name by multiples -- its AI-driven repricing since 2023 is an outlier event that inflates the basket's apparent quality. Tesla and Google trail significantly, revealing that passive Mag 7 exposure is effectively a concentrated NVDA + MSFT + AAPL bet, not a diversified tech allocation.

Chart B: Annual Return Heatmap by Stock and Year



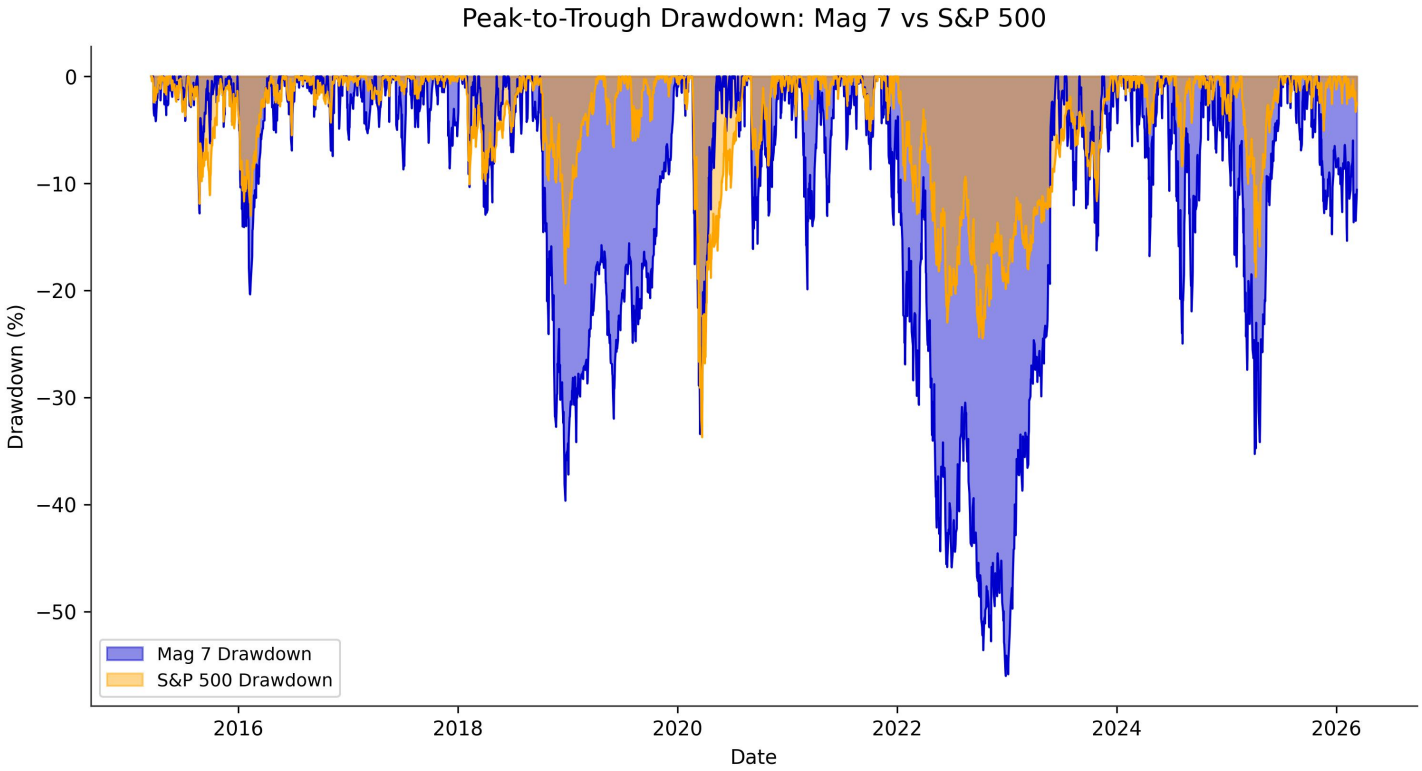
The calendar breakdown confirms regime-dependency. In 2022 -- the rate shock year -- every Mag 7 stock posted deep negative returns. The 2023-24 recovery was led by NVDA (AI premium), Meta (efficiency reset), and MSFT (Azure AI). Tesla has been the most volatile name year-on-year, with swings of +100% (2020) to -65% (2022), making it the basket's highest-risk constituent.

Chart C: Rolling 12-Month Sharpe Ratio -- Mag 7 vs S&P 493



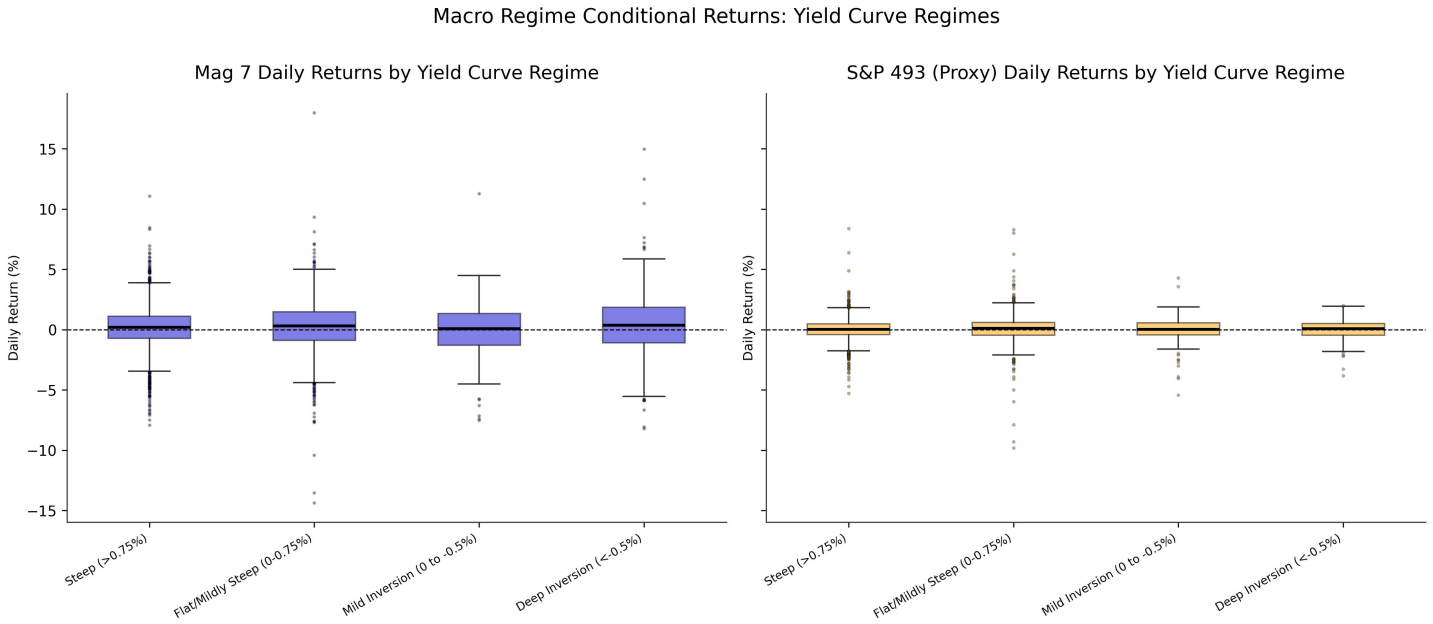
The Sharpe premium of Mag 7 over the broader S&P 493 has not been constant. It collapsed entirely during the 2022 rate hike cycle -- when Mag 7 Sharpe turned sharply negative while the S&P 493 held up better on a risk-adjusted basis. This is the clearest evidence that Mag 7 is duration-sensitive growth risk, not a structural quality premium. The Sharpe recovered only on the back of the AI boom, driven by a single stock.

Chart D: Peak-to-Trough Drawdown -- Mag 7 vs S&P 500



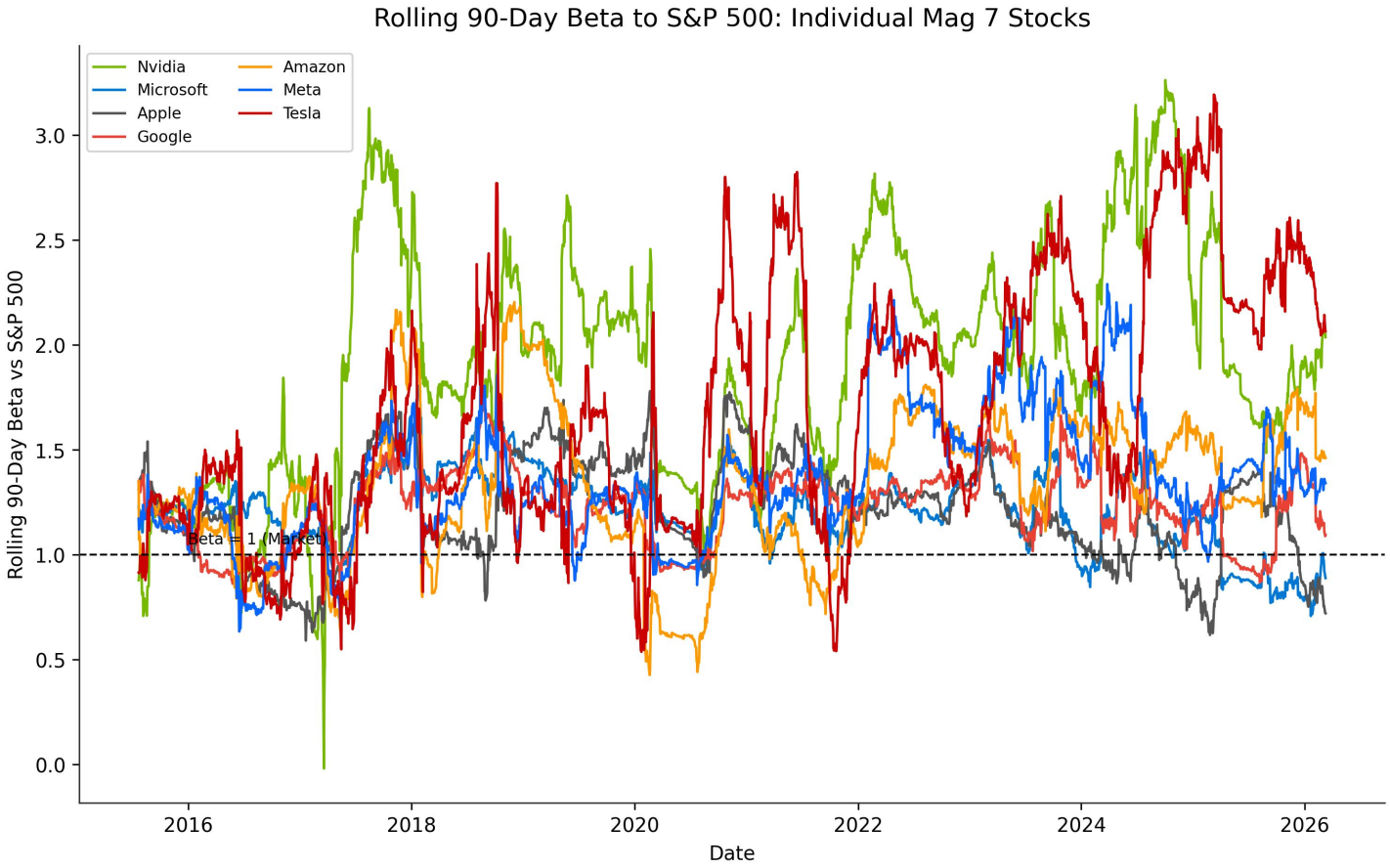
The Mag 7 basket sustains far deeper drawdowns than the broader S&P 500. The COVID crash in March 2020 and the 2022 rate cycle each produced Mag 7 drawdowns exceeding -40%, versus -34% for the S&P 500. Crucially, Mag 7 takes longer to recover. A risk-parity investor allocating more to Mag 7 for "quality" is actually accepting significantly worse tail-risk than a broad-market position.

Chart E: Macro Regime Conditional Return Distributions



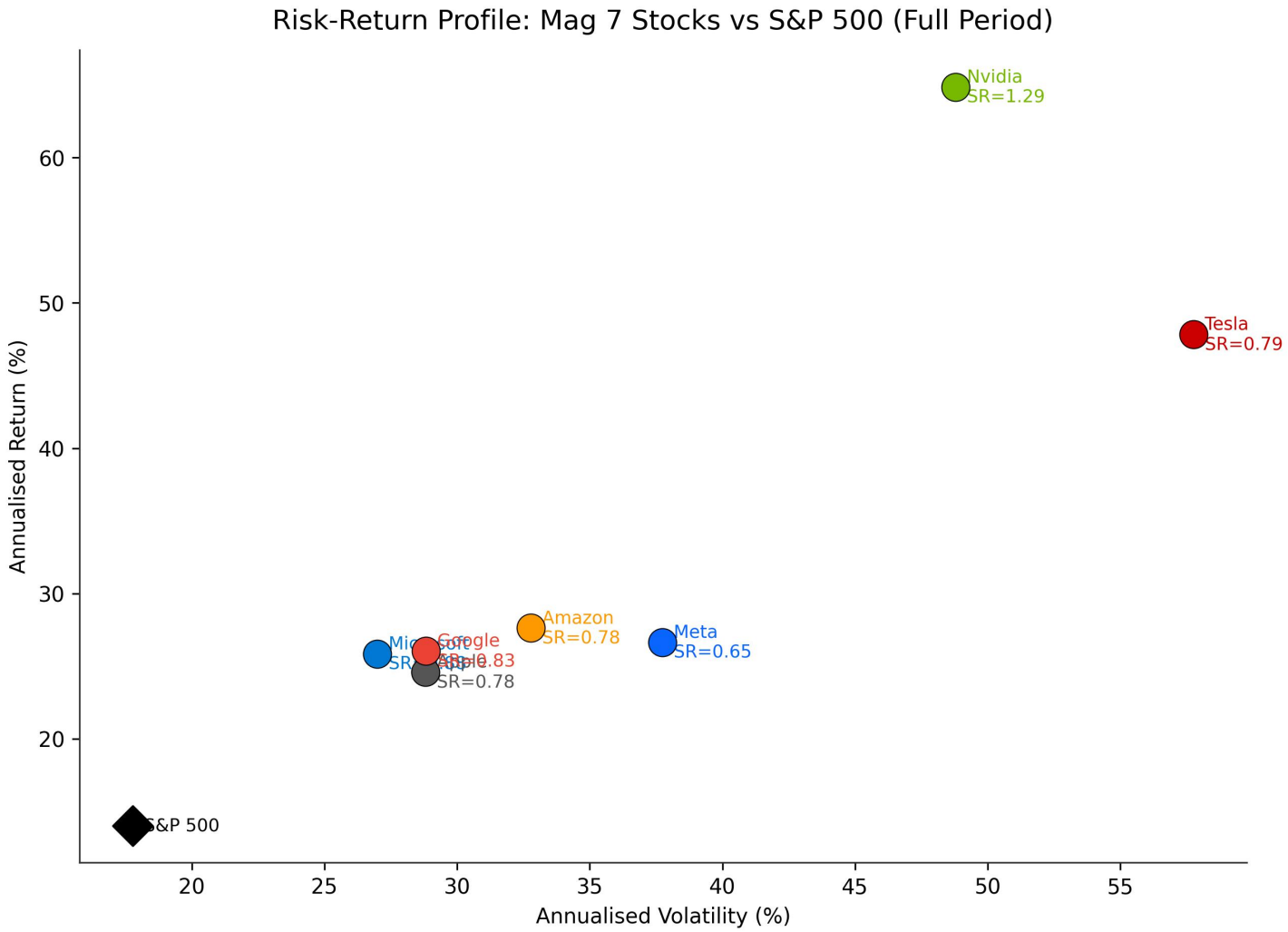
Conditioning returns on the yield curve regime reveals a clear pattern. Mag 7 returns are highest in steep yield curve environments (positive spread > 0.75%) and worst during deep inversions -- consistent with long-duration growth equity behaviour. The S&P 493 shows far less sensitivity to the yield regime, confirming that concentration in Mag 7 introduces a macro rate bet that is absent from more diversified exposure.

Chart F: Rolling 90-Day Beta to S&P 500 -- Per Stock



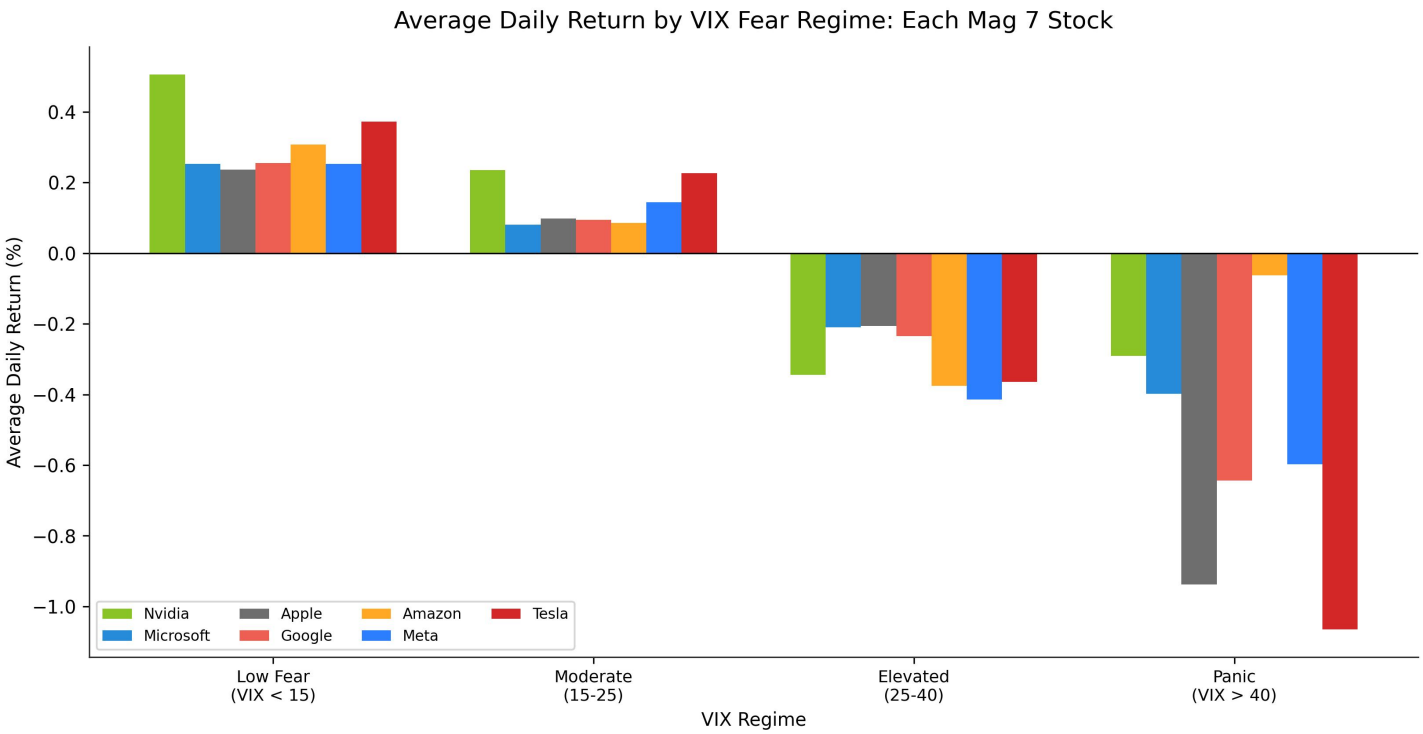
Betas across the Mag 7 have been far from static. Tesla exhibits the most extreme beta oscillations (ranging from below 0 to above 3x), reflecting its dual nature as both a consumer brand and a speculative vehicle. Nvidia's beta surged post-2023 as its AI correlation decoupled from the broader market. Apple remains the closest to beta = 1, justifying its role as the basket's defensive anchor. Microsoft's beta has compressed as it repriced as a hybrid growth/dividend compounder.

Chart G: Risk-Return Scatter -- Efficient Frontier View



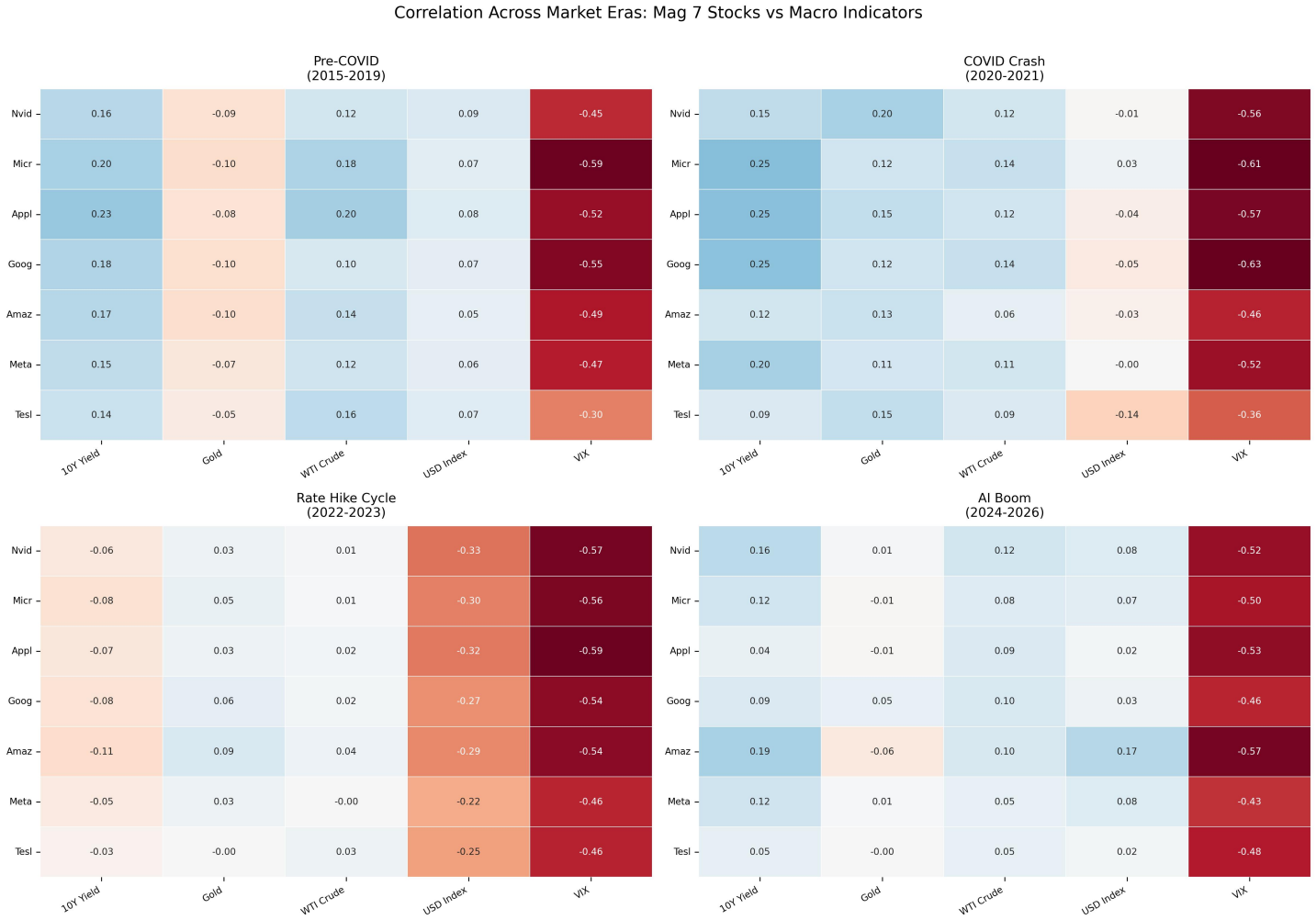
On a full-period risk-return basis, Nvidia is the sole clear outlier -- exceptional return for its level of volatility. Apple and Microsoft plot near the efficient frontier edge: good returns for controlled risk. Tesla sits in the worst quadrant: high volatility, mediocre long-run return -- a net drag on the basket's Sharpe. The S&P 500 (diamond marker) sits in the middle, confirming that most Mag 7 stocks individually do NOT dominate a diversified benchmark on a risk-adjusted basis.

Chart H: Average Daily Return by VIX Fear Regime



Segmenting returns by VIX bucket reveals a consistent pattern: as market fear rises, Mag 7 returns collapse more sharply than the blended index. During Panic regimes (VIX > 40), every Mag 7 stock posts negative average daily returns, with Tesla and Nvidia showing the sharpest losses. This confirms that the Mag 7 basket provides zero defensive protection in true market stress -- it amplifies drawdowns rather than cushioning them.

Chart I: Correlation Shifts Across Four Market Eras

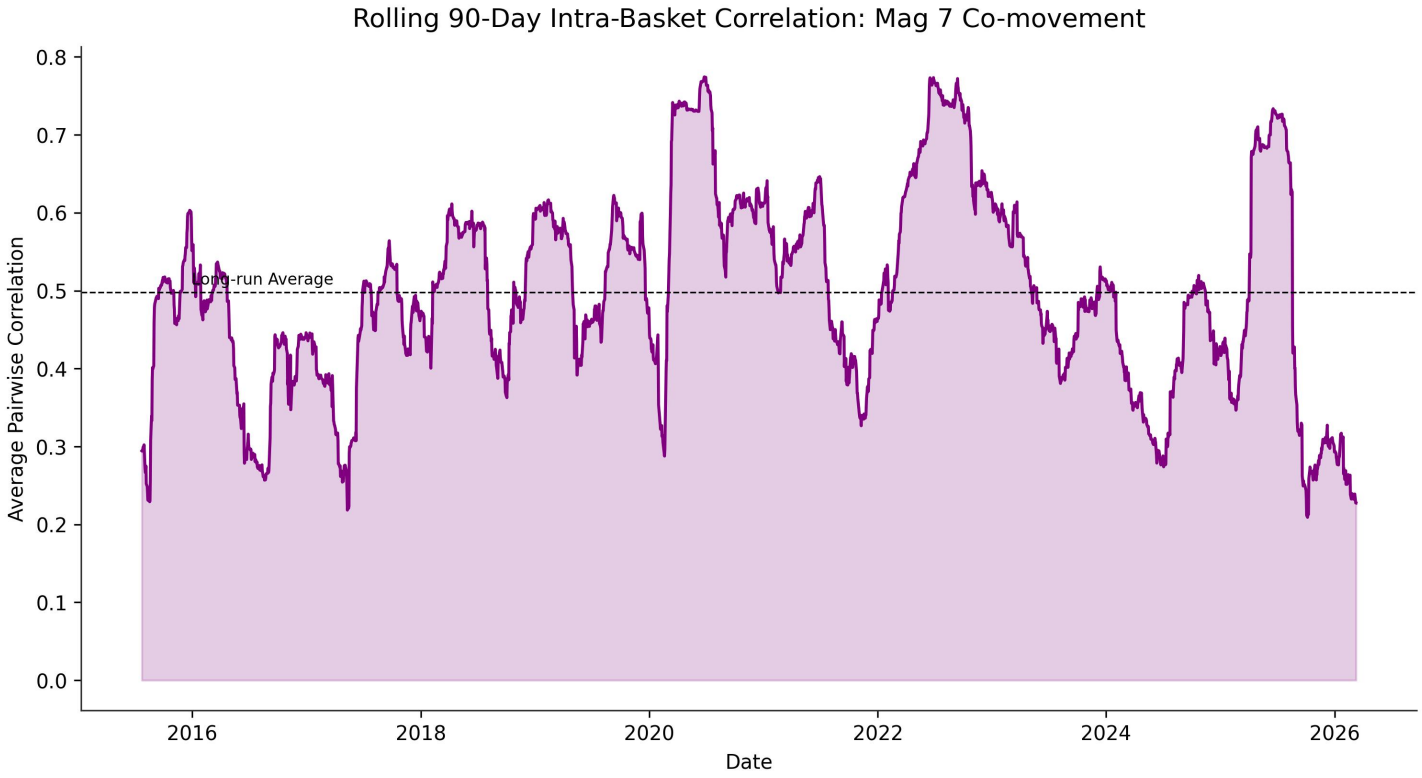


Perhaps the most important chart in this report. Correlations between Mag 7 stocks and macro indicators are highly regime-dependent. In the Pre-COVID era, most Mag 7 names were weakly correlated to rates and commodities. Through the Rate Hike Cycle (2022-23), correlations to 10Y yields turned sharply negative for growth names. In the AI Boom era (2024+), correlations decoupled -- NVDA began tracking VIX more than yields, reflecting speculative positioning. This instability makes static correlation-based hedging dangerously unreliable.

Chart J: Per-Stock Yield Sensitivity Regression

The regression slopes confirm the rate-sensitivity hierarchy. Tesla and Meta show the steepest negative slopes -- a 0.1% jump in 10Y yields typically corresponds to a -0.3% to -0.5% daily return. Apple and Microsoft are the least sensitive. Nvidia is anomalous: its slope has been weakly negative because AI narrative demand often overrides rate-driven discount-rate logic. This analysis gives portfolio managers a precise duration-proxy measure for each name.

Chart K: Rolling Intra-Basket Co-movement



When Mag 7 stocks all move together (high average pairwise correlation), the basket offers no diversification benefit -- it functions as a single factor bet. Intra-basket correlation spiked to near 0.9 during COVID and the 2022 crash, periods when diversification was most needed. Post-2023, correlation declined as the AI-driven bifurcation between NVDA winners and others widened. A low intra-basket correlation today suggests the basket may no longer be a coherent investment theme.

Conclusions and Portfolio Implications

1. The Basket is Internally Incoherent.

Nvidia's AI-driven repricing has decoupled from the rest of the Mag 7. The basket is effectively a single-stock bet masquerading as diversification.

2. The Sharpe Premium is Cyclical, Not Structural.

Mag 7 Sharpe superiority disappears during rate-tightening cycles. A buy-and-hold Mag 7 strategy implicitly bets on perpetual monetary accommodation.

3. Drawdowns Are Deeper and Longer Than the Benchmark.

At every major stress event, Mag 7 lost more and recovered slower than the S&P 500. Risk-parity arguments for Mag 7 overweight are empirically unsupported.

4. The Correlation Regime Has Changed.

Pre-2020 correlations to macro variables are no longer valid. A hedge calibrated on 2015-2019 data would have been completely ineffective in 2022-2024. Dynamic hedging is essential for any Mag 7 concentration.

5. Tesla Belongs in a Different Risk Bucket.

Its beta oscillations, drawdown depth, and VIX sensitivity are categorically different from the other 6 names. It is a macro/sentiment vehicle, not a tech compounder.

6. Recommended Analytical Framework.

Treat the Mag 7 as 3 sub-groups: (a) AI Infrastructure -- NVDA, standalone risk; (b) Compounders -- MSFT, AAPL, GOOGL, suitable for long-term quality allocation; (c) High-Beta Cyclicals -- TSLA, META, AMZN, treat as risk-on expressions.